What this writing is, and what it costs

The gain slope is read off the GAIN STATE rather than off a height: $\bar{a}' = b(1-\bar{a})^2$ evaluated at $\hat{\bar{a}}_w$. That is the whole difference from the height writing, and everything below follows from it.

The price is structural, not a tuning failure. Because $\bar{a}'$ depends on $\hat{\bar{a}}_w$, the error recursion gains the term $-2\hat{b}(1-\hat{\bar{a}}_w)M$, so the homogeneous coefficient exceeds 1 while the stage descends. Over the canonical descent that factor accumulates to 220×, and that number is not an artefact: the true slope at the trough is 107× the slope at the seed, so the writing converts a real property of the physics into error growth. The height writing's coefficient for that path is exactly zero.

The filter prices it correctly and can do nothing else. $F_e(4,4)$ carries the term, so P_{44} grows 1.8× during motion, L grows 5.6× on y_1 and 23–39× on y_2, and the estimate leans on measurements that are unbiased and noisy. The across-seed spread is 6.4× the exogenous arm's. That is the bias-for-scatter trade, bought deliberately by an optimal filter facing a model it should not trust.

Demonstrated with a deterministic injection: seeding b at 1 instead of 8/9 puts $\hat{\bar{a}}_w[0]$ a known +0.0056 above the truth, identical across all eight seeds and carrying no random information. Fed $\bar{a}'$ from the true height the offset is erased and the across-seed band stays narrow; fed $\bar{a}'$ from the belief it is not erased and the band opens out. A common-mode model error becomes seed-dependent scatter, and the split appears only once the stage moves. Figure `formC_belief_vs_command.png`.

What the reparameterisation bought

b multiplies rather than divides (user decision, 2026-08-18). Same curve family, $b_{\text{new}} = 1/b_{\text{old}}$, so the far-field limit inverts $9/8 \to 8/9$ while contact stays exactly 1.

Every known truncation is unchanged in magnitude, for one structural reason: each is a statement about the product $\bar{a}' = b(1-\bar{a})^2$, and that product is the reparameterisation invariant. Even the relative prior width is exactly invariant, since half/mid $= (hi - lo)/(hi + lo)$ is fixed under $x \mapsto 1/x$: 1.21052 % either way.

One thing is strictly better and had not been written down. In the old form $\bar{a}' = (1-\bar{a})^2/b$ is nonlinear in b, so $\partial^2\bar{a}'/\partial b^2 = 2(1-\bar{a})^2/b^3$ and a curvature term was dropped from $F_e(4,5)$ and $H(2,5)$ every step, worth $e_b/\hat{b} = 1.21\,\%$ at the one-sigma prior. In the new form $\bar{a}'$ is exactly linear in b and that term is identically zero: the b column is now exact.

S1 — the truth, and what the model replaces

$\bar{a}_{\text{true}}$ is $D_\perp$ from `calc_correction_functions` character for character, since $\bar{a} := a/a_o = 1/c_\perp$ and $c_\perp = 1/D_\perp$.

Because $\bar{a}_{\text{true}}$ is monotone in $\bar{w}$, $\bar{w}$ can be eliminated EXACTLY and the autonomous law $\bar{a}' = b_{\text{true}}(\bar{a})(1-\bar{a})^2$ carries no approximation. The model's single approximation is replacing the function $b_{\text{true}}(\bar{a})$ by a constant. Stating it this way also dissolves the apparent conflict between the two anchors: b_{true} is a curve, its ends simply read differently, and choosing an end is a design decision.

Verified at 30 digits: $\bar{a}_{\text{true}}(1) = 0$ to 5×10^{-32}, $\bar{a}'_{\text{true}}(1) = 1$ exactly, so $b_{\text{true}}(\bar{a} = 0) = 1$; and $b_{\text{true}} \to 8/9$ FROM BELOW.

$b_{\rm true}$ is NOT monotone. It rises from an interior minimum 0.866978 at $\bar{w} = 2.193$ to 0.888225 at the envelope top and never reaches 8/9. The envelope range is therefore $[0.866978,\ 0.888225]$, whose lower end is an interior extremum rather than an endpoint.

Seed and prior — the seed and the width come from different things

$$\hat{b}[0] = \tfrac{8}{9}, \qquad \sqrt{P_{55}[0]} = \sup_{\bar{w}\in\mathcal{E}} \left| b_{\rm true}(\bar{w}) - \tfrac{8}{9} \right| = 0.021911$$

THE SEED is the analytic anchor because $P[0]$ is the belief at $t = 0$ and the run starts in the far field. At the seed height $\bar{w} = 22.22$ the truth is $b_{\rm true} = 0.888167$; the anchor misses it by 7.2×10^{-4}, the envelope midpoint 0.877602 by 1.06×10^{-2} — fifteen times worse, because the midpoint is dragged down by an interior minimum the run reaches twenty radii later. Measured over the eight house seeds, seeding at the midpoint costs +1.81 points of overall RMS, sd 1.52, SEM 0.54, $t = +3.38$, worse on 8/8. This is the same defect as sizing floor$_a$ from the envelope supremum, which cost 2.9 points and was repaired the same day: both take a statistic over ground the run has not covered and use it to set a belief at $t = 0$.

THE WIDTH is a statement about the run, not about $t = 0$. b is a constant state ($b[k + 1] = b[k]$, $Q_{55} = 0$), so one number must serve the whole traverse including the near wall. That is step 4 of the $P[0]$ chain in `reference/shared/param_prior_rules.md`.

KNOWN CONSEQUENCE, recorded rather than repaired. The shape-acceptance criterion compares $\sup|\theta_{\rm eff} - \theta_0|$ against $\sqrt{P[0]}$, and with $\theta_0 = 8/9$ those are the same supremum, so the ratio is identically 1.0000 and the test cannot fail. That is a defect in the TEST's denominator. Distorting the prior to give the test something to do is what mis-seeded the filter, so it is not done. Related: the tool the criterion names, `verify_shape_exponent_bound.m`, DOES NOT EXIST in this repository, so no member of this family has been through it. That is the missing acceptance artifact.

CHARTER TENSION. $\mathcal{E}$ is built from `cfg.h_bottom` and `cfg.h_init`, so $\sqrt{P_{55}[0]}$ is trajectory-dependent, while CLAUDE.md describes this family's prior as global and trajectory-independent. The anchor form of the seed is global; the width is not.

CORRECTION TO AN EARLIER READING. $b[k + 1] = b[k]$ with $Q_{55} = 0$ alongside S1's varying $b_{\rm true}$ is NOT a self-contradiction. It is the six-step $P[0]$ chain executed correctly, with the prior serving as the container for the variation. What was genuinely missing was this citation.

Truncations written with "=" — seven, none previously declared

Magnitudes are on the production scenario, $R = 2.25\,\mu$m, $\Delta t = 1/1600$, $\lambda_c = 0.7$, 1 Hz $\times$ 2.5 μm about $h = 7\,\mu$m, $\mathcal{E} = [1.9,\ 23.222]$.

1. S3's $\bar{a}_w[k + 1] = \bar{a}_w + \bar{a}'_{\rm true}\Sigma$ is forward Euler for $\bar{a}(\bar{w} + \Sigma)$. Dropped: $\frac{1}{2}\bar{a}''\Sigma^2$. Per-step relative error $b(1 - \bar{a})|\Sigma| = 0.200\,\%$ at the trough, $0.133\,\%$ over the cycle; accumulated $+5.90 \times 10^{-4}$ per oscillation cycle. $\bar{a}'' < 0$ everywhere and the term goes as Σ^2, so it does NOT change sign with direction of travel: a closed oscillation does not cancel it and the estimator runs high. This is the project's quadrature ratchet.

2. S6's $e_{\bar{a}_w}$ recursion is written as an identity but drops $+\hat{b}\, e^2 M$, which no matrix F_e can carry. Size against the retained $-2\hat{b}(1-\hat{\bar{a}}_w)eM$ is $e/(2(1-\hat{\bar{a}}_w)) \approx 0.31\,\%$. The same expansion also drops the cross term $-2(1-\hat{\bar{a}}_w)e_{\bar{a}}e_b M$, about 1.25 % of the retained e_b column.

3. S7's back-off uses the COMMANDED $\nabla_d \bar{w}_d$; the actual d-step change is $\nabla_d \bar{w}_d - \delta\bar{w}_3[k] + \delta\bar{w}_1[k]$. So $H(2,1) = -\bar{a}'$ and $H(2,3) = +\bar{a}'$ are MISSING COLUMNS, not small coefficients. At this project's reported z tracking errors they are 25–50 % of the retained term, and $H(2,1)$ puts y_2 on the same column y_1 measures — the channel the y_1-versus-y_2 decomposition found dominant.

4. The same back-off is itself forward Euler over a $d = 2$ gap, carrying four times the single-step quadrature error: $+8.56 \times 10^{-6}$ per application at the trough, landing as a one-signed bias on the y_2 innovation rather than integrating into the state. Independent of item 3; fixing one does not fix the other.

5. S8's $q_3 = -\bar{a}_w \bar{f}_T$ drops the MA(2) memory and the $(1-\lambda_c)n_w$ share of ε_w that S3 itself defines. $\mathrm{Var}(\varepsilon_w)$ is understated $1.18\times$ and its DC power $2.17\times$ — the same factor the formB MA(2) work landed on. HERE THE CODE IS RIGHT AND THIS DOCUMENT IS WRONG: the controller carries the full variance.

6. $P_{44}[0]$ adds the $\bar{w}_s$ and b contributions as if the two priors were independent, while $P_{45}[0]$ on the next line asserts a nonzero correlation between b and $\bar{a}$. The independence is undeclared.

7. S8 mixes $\bar{a}_w$ and $\hat{\bar{a}}_w$: Q_{33}, R_2 and $\sigma^2_{\delta\bar{w}_r}$ are written at the TRUE gain while Q_{34} and Q_{44} use the estimate and the code uses the estimate throughout. For a document whose charter constraint is run-time c-free evaluation at the estimate, that is not a leavable typo.

An assumption violation the truncations hide

ε_w contains $(1-\lambda_c)n_w[k]$ and y_1's noise IS $n_w[k]$, so $E[\mathbf{q}\mathbf{v}^\top] \neq 0$, which the standard Kalman filter this document invokes forbids, and $R = \mathrm{diag}(R_1, R_2)$ carries no cross term. Coefficient 0.0157. Small, but structural rather than approximate — and invisible inside S8's own truncated q_3, which drops the n_w share. The truncation hides the violation, which is why it survived.

Domain, and what the file does not contain

$\bar{a}(\bar{w}) = 1 - 1/[b(\bar{w} - \bar{w}_s)]$ has a pole at $\bar{w}_s$ and is NEGATIVE for $\bar{w} - \bar{w}_s < 1/b \approx 1.125$, contradicting the main file's own $0 < \bar{a} < 1$. The condition $\bar{w} - \bar{w}_s > 1/b$ is stated nowhere; production carries it as `GAP_FLOOR` and the $\bar{a}$ clamps. Separately, $0 < \bar{a} < 1$ is open at 0 while S1's contact anchor is stated AT $\bar{a} = 0$.

P^f appears once, in $L[k]$, and is never defined. Missing with it: the covariance prediction, the covariance update, the state update — so $L[k]$ is computed and nothing consumes it — and the predicted measurement. The last is a trap rather than an omission: $\hat{y}_2$ is NONLINEAR, $\hat{y}_2 = \hat{\bar{a}}_w - \hat{b}(1-\hat{\bar{a}}_w)^2 \nabla_d \bar{w}_d$, not $H\hat{\mathbf{x}}$, and production carries an explicit warning that using H there is wrong. As the file stands the only available reading is the linear one.

Also absent: Q is never assembled ($Q_{11} = Q_{22} = 0$, Q_{35}, Q_{45} never stated); S4 is titled "Process model" but contains only the state vector; the MA(2) augmentation production runs by default has no representation at all, so the file is behind production in state dimension as well.

Consider b — the law's shape error carried, not estimated (2026-08-27)

What the main file changed on 2026-08-27, and why. Up to that date b was a state: $b[k+1] = b[k]$, L_5 free, P_{55} shrinking. Three facts, all measured on the deep band, say that construction cannot do the job it was given.

First, b_{true} is not a constant along the path. From the two-sphere series, $b_{\mathrm{true}}(\bar{a}) - 8/9$ is 0 in the far field, -0.022 at $\bar{a} = 0.51$, $+0.039$ at the trough $\bar{w} = 1.10$ and $+0.111$ at contact — it crosses zero at $\bar{a} \approx 0.23$. The state b can hold ONE value; the truth it stands for is a curve. Second, b is barely observable: both of its Jacobian columns ($F_{e,45}$ and H_{25}) are proportional to the displacement, both vanish in a hold, and the CRLB over the whole scenario only matches the prior (observability ledger, 2026-08-17). Letting it move is letting an unobservable parameter be dragged by whatever innovation arrives: widening its prior put $\hat{b}$ 300σ from the truth while P_{55} shrank $512\times$ (2026-08-20). Third, and decisive: locking b at 8/9 versus estimating it changes the trough answer by $+0.10 \pm 0.14\,\%$ (8 seeds, paired, 2026-08-26). Estimating b buys nothing, and while it is being estimated P_{55} shrinks by a third, so the ONE thing the b slot could still contribute — its uncertainty — is being thrown away.

That uncertainty is exactly what the filter is missing. The injection-response test (2026-08-26, `run_inject_response.m`) put a known $\pm 5\,\%$ error into $\hat{\bar{a}}_w$ mid-descent and watched it grow to $2.4\times$ before y_2 pulled it back near the wall. Decomposing the y_1 leg showed the growth comes through the innovation, and three discriminating arms located its sign in the competition between $F_{e,34}P_{44}$ (a gain error moves the particle; correcting) and $Q_{34} = -\bar{a}'Q_{33}$ (a thermal kick moves particle and gain together; not correcting): with P_{44} built from $Q_{44} = \bar{a}'^2 Q_{33}$ alone the second term wins wherever the applied force is small and $\bar{a}'$ is large, and a control-induced lag is read as a thermal kick. P_{44} carries the thermal wander of the gain and nothing else; the law's own error has no container. Inflating Q_{44} uniformly moved the canonical trough bias from $+22.65\,\%$ to $+6.89\,\%$ ($t = -5.1$) and proved the lever, but also inflated P_{44} in the holds where the law is not integrating at all.

The container that has the right shape is the one the file already owns. The missing forcing on row 4 is $e_b[k]\,(1-\hat{\bar{a}}_w)^2[\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3]$ with $e_b[k] = b_{\mathrm{true}}(\bar{a}_w[k]) - \hat{b}[0]$, bounded by $\sqrt{P_{bb}}$, and $F_{e,45}$ is precisely its coefficient. Keeping $P_{55} \equiv P_{bb}$ while zeroing L_5 — a Schmidt–Kalman consider parameter — makes the covariance propagation carry e_b's effect on $\bar{a}_w$ every step without pretending to learn e_b. The identity closing the Seed-and-prior section is the estimator's own recursion read backwards: $(1-\hat{\bar{a}}_w)^2[\cdots] = \Delta\hat{\bar{a}}_w/\hat{b}$ between one posterior and the next prior, so the sum of $F_{e,45}$ telescopes to the gain integrated so far over $\hat{b}$, minus the measurement corrections $\sum L_{4,\cdot}\nu$ that the recursion does not see (exact, both terms written). Added to the seed $P_{45}[0] = (1-\hat{\bar{a}}_w[0])P_{bb}/\hat{b}$ that the file sets eight lines earlier, the first-order cross term is $(1-\hat{\bar{a}}_w[k])P_{bb}/\hat{b}$ and the induced P_{44} term is $[(1-\hat{\bar{a}}_w[k])/\hat{b}]^2 P_{bb}$: the static sensitivity $e_{\bar{a}}|_b = e_b(1-\bar{a})/b$, which the estimating form applied at $k=0$ and then let shrink, held at every k. At the canonical trough that is 0.040 against a filter $\sqrt{P_{44}}$ of 0.0073; in the far field 0.0022. It is a function of the current estimate only, so it unwinds on the way back up, is identical at the two ends of a closed oscillation cycle, is constant in a hold, needs no accumulator state, and reads nothing of $c(\bar{w})$ at run time. It is an upper bound: $b_{\mathrm{true}} - 8/9$ changes sign along the descent and partly cancels (the true integrated law error on the canonical descent is $+0.0085$, a quarter of the bound), so the honesty ratio it will produce is expected below one. A container that carries the sign of $b_{\mathrm{true}} - 8/9$ would need the second coefficient of the reflection series; that is a separate decision and is not made here.

Three things declared. The $\approx$ in that block neglects $F_{e,44} \neq 1$, the $F_{e,43}P_{35}$ coupling and the measurement updates of P_{45} and P_{44}, so the closed form is the first-order shape, not the value the filter carries. P_{55} itself is never updated because $L_5 = 0$, which is what makes $\equiv P_{bb}$ exact. And row 5 of F_e with $q_5 = 0$ is the CONSIDER MODEL, not the truth: the true $e_b[k]$ drifts along the path by $b_{\text{true}}(\bar{a}_w[k+1]) - b_{\text{true}}(\bar{a}_w[k])$, and that drift is not propagated — it is covered by the bound $|e_b| \le \sqrt{P_{bb}}$, which is why P_{bb} must be the supremum over the envelope and not the width of a constant. (Independent review, 2026-08-27: items 1–3, 6, 8 confirmed; the seed term and the "$\supseteq$" were its two corrections.)

What must be shown before this replaces the estimating form as production: with the consider flag on and the hooks off, the seed-7 fixture must still reproduce when the flag is off; on the Meng ramp the injected error must decay instead of growing and $L_{1,4}$ must turn non-negative on the crossings; on the canonical band the trough bias must fall to the neighbourhood of the uniform $\times 3$ arm while the TOTAL honesty $\sqrt{\text{bias}^2 + \text{spread}^2}/\sqrt{P_{44}}$ — not spread alone, since the forcing is deterministic — stays within $[0.3, 1.2]$ in the holds; and the result must not move under $a_{\text{cov}} \times 2$ and $\div 2$.

The law error as process noise — the correlated container (2026-08-27)

Consider-b was run before this section was written, and it is why this section exists. On the canonical band it moved the trough bias from $+22.65\,\%$ to $+10.09\,\%$ (paired $t = -5.0$) and the total honesty from 2.87 to 1.73, but $\sqrt{P_{44}}$ at the trough reached 0.0087 against the first-order 0.040: the correlation P_{45} that carries e_b into P_{44} is consumed by the y_1 updates, $P^+_{45} = P^-_{45} - L_{4,\cdot}(HP^-)_{\cdot 5}$, and those are the very updates whose sign is the defect. A one-time correlation can be learned away by the wrong information; the Meng injection response with consider-b on was unchanged (peak 2.1 against 2.4). The container has to re-enter every step.

What re-enters is the missing forcing $e_b[k]\,g[k]$ with $g = F_{e,45}$. On a monotone command segment e_b is one-signed (measured 2026-08-27: the true $b_{\text{true}} - 8/9$ changes sign twice over the whole canonical descent, and the accumulated forcing has correlation factor $|\sum e|/\sqrt{\sum e^2} = 12.7$ there, 59.6 on the Meng ramp), so within the segment it is an unknown constant of magnitude at most $\sqrt{P_{bb}}$ and the accumulated forcing is bounded by $\sqrt{P_{bb}}\,|G[k]|$ with G the running sum of g. A per-step injection whose running sum is $P_{bb}G^2$ is $q_{\text{law}} = P_{bb}\,g\,(2G_{k-1} + g)$: non-negative because g and G share a sign on the segment, proportional to the running correlation $2G/g$ rather than to g^2, zero in a hold, and restarted from zero at every command turning point so an oscillation cannot accumulate a bound the truth does not have. Everything in it is either the filter's own Jacobian or the command increment; $c(\bar{w})$ is not read.

Its size, from the telescoping identity: at the end of a segment $\sqrt{\sum q_{\text{law}}} = \sqrt{P_{bb}}\,|\hat{\bar{a}}_w[k] - \hat{\bar{a}}_w[j_0]|/\hat{b}$, i.e. 0.038 over the canonical descent and 0.026 per oscillation half-cycle; per step it is $4\times$ the thermal Q_{44} at $\bar{a} = 0.5$ and $11\times$ at $\bar{a} = 0.1$, and exactly zero in the holds. The uniform $Q_{44} \times 3$ and $\times 10$ arms of 2026-08-26 injected the same order per step on the descent and reached bias $+6.9\,\%$ / $+8.1\,\%$ with total honesty 0.82 / 0.80; they are this container without its shape.

Declared: the one-signed bound is an upper bound (the true integrated law error on the canonical descent is $+0.0085$ against the bound 0.038, because $b_{\text{true}} - 8/9$ crosses zero at $\bar{a} \approx 0.23$); q_{law} is a bookkeeping increment and is kept out of the $d\,Q_{44}$ delay term of R_2, which is 10^{-7} of R_2 either way; and the segment is defined on the COMMAND increment, so the small tracking-error share of g inside a hold is counted as zero.

Registered before the run (canonical deep, 8 seeds, arm best with the flag): trough bias in $[+5, +10]\,\%$; total honesty in the end hold in $[0.8, 1.5]$; per-seed sd $\le 12\,\%$; descent $L_{1,4} \ge +0.07$; on the Meng ramp the injected error peaks below 1.5 and decays with a non-positive y_1 ledger; $a_{\mathrm{cov}} \times 2$ and $\div 2$ move bias by < 1 pp and honesty by < 0.1; flag off reproduces the fixture and the shallow scenario bit for bit. If total honesty comes out below 0.5 the bound is too loose and the sign-carrying container (second reflection coefficient) is next; if the bias stays above 12 % with honesty near one, the container is honest and the bias is not the law's.

Results

Canonical scenario, 8 house seeds, paired, overall relative gain-error RMS. All three arms use the seed-local floor$_a$ and the anchored b prior.

	desc peak	osc RMS	overall	$\hat{b}$[end]
b fixed at 8/9	4.78 ± 3.27	1.68 ± 0.82	2.20 ± 1.33	0.88889
b estimated	5.33 ± 3.39	1.96 ± 1.01	2.46 ± 1.58	0.89218 ± 0.01096
$\bar{a}'$ from the TRUE height	1.21 ± 0.32	0.68 ± 0.47	1.16 ± 0.52	0.88752 ± 0.01505

Estimating b costs +0.258, sd 0.280, SEM 0.099, $t = +2.61$, better on 1/8. Consistently a little worse: b receives almost no information, so a free b is a locked b plus a noise source. The ceiling with a perfect integrand is 1.16 % against the 2.20 % achieved, and that gap is the price of reading $\bar{a}'$ at the belief.

TRACKING ERROR, the fourth figure row, is $R\,\delta\bar{w}_3$ in μm. With the sample alignment corrected the bias is zero in every phase — -0.74 nm on the descent, -1.04 ± 3.29 nm over the eight seeds in the initial hold, $t = -0.89$ — and the row carries noise only, RMS 25.08 nm against a 3.31 nm measurement noise spec.

Verification record

Thirteen relations CAS-checked, then an independent pass swept every occurrence of b cell by cell including the nested arrays inside F_e, re-derived S6's recursion from scratch, and reconfirmed both limits at 30 digits. No missed or half-converted b; the five surviving denominators are all in the seed and all correct for the multiplicative form.

The implementation was checked against the derivation by an IDENTITY rather than by agreement of numbers. $b_{\mathrm{new}} = 1/b_{\mathrm{old}}$ is exact, so a run locked at $b_{\mathrm{old}} = 1$ must reproduce one locked at $b_{\mathrm{new}} = 1$, and 9/8 must reproduce 8/9. Measured: $\hat{\bar{a}}_w$ identical to 0 and to 6.1×10^{-16} respectively. Only all eight conversion sites being right produces that.

Open items

1. The seven truncations above are now declared but not repaired. Items 3 and 5 are the two where the document is wrong rather than merely silent.
2. P^f and the four missing filter lines. The nonlinear $\hat{y}_2$ is the one that changes what a reader would build.
3. `verify_shape_exponent_bound.m` does not exist, so the shape acceptance this family's charter names has never been run on any member.

4. The acceptance criterion's denominator is the same supremum as its numerator when θ_0 is the anchor. The criterion needs a different denominator, not a distorted prior.

5. Production comments referencing "S5(b)" point at a 7-state version that was never written and whose header claim has been withdrawn.